

Explainable Machine Learning (XAI) for Anomaly Detection in Medical IoT Sensor Data

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Abstract—Medical Internet of Things (mIoT) devices have revolutionised modern healthcare by enabling the continuous, real-time monitoring of critical physiological parameters, including heart rate, oxygen saturation, and respiratory rate. The rapid integration of Medical IoT devices into clinical workflows has generated an unprecedented volume of high-velocity physiological data. The primary aim of this research is to design, implement, and evaluate an explainable anomaly detection framework specifically optimized for Medical IoT telemetry. Unlike existing black-box models that offer binary “normal/abnormal” outputs, the proposed system integrates Autoencoder-SHAP pipeline and LIME into a multi-model anomaly detector pipeline. The framework aims to mitigate alarm fatigue and foster the trust necessary for the clinical adoption of AI-driven patient monitoring systems. The findings confirm that explainability is not optional in healthcare AI systems but a fundamental requirement for safe deployment.

I. INTRODUCTION

A. Background

The emergence of the Medical Internet of Things (mIoT) has transformed patient care by enabling continuous monitoring of key physiological signals such as heart rate, oxygen saturation (SpO_2), and respiratory patterns [1], [2]. These high-frequency data streams create strong opportunities for machine learning (ML) models to detect early signs of clinical deterioration [3]. However, a significant “interpretability gap” remains [4]. Although advanced models such as Isolation Forests and Autoencoders can effectively detect anomalies, they typically operate as “black boxes,” obscuring the physiological reasoning behind alerts [5], [6]. This lack of transparency limits clinical trust and real-world deployment [7].

In critical care environments, interpretability and accountability are essential [8]. Opaque systems can contribute to “alarm fatigue,” where frequent, non-actionable alerts desensitise clinicians and increase the risk of missed critical events [9]. While Explainable AI (XAI) methods such as SHAP and LIME have shown effectiveness in static settings, their use in dynamic, noisy, and unsupervised mIoT data remains limited [10], [11]. This study addresses this gap by developing a “glass-box” anomaly detection framework that combines accurate detection with clinically meaningful explanations.

B. Problem Statement

The integration of mIoT devices into clinical workflows has generated large volumes of high-frequency telemetry data [12]. Although unsupervised models ranging from tree-based methods to deep reconstruction approaches can detect anomalies, they often lack the transparency required for real-time

clinical use [13]. This limitation leads to two key issues. First, clinicians cannot reliably distinguish between true physiological deterioration and technical artifacts such as sensor noise or device failure [14]. Second, opaque systems contribute to “alarm fatigue,” where excessive, non-interpretable alerts reduce clinician responsiveness and compromise patient safety [15].

In addition, existing XAI methods face challenges in handling noisy time-series data and operating under the computational constraints of mIoT edge environments [16]. Without clear feature-level explanations, these systems lack the “clinical coherence” needed for trust and regulatory acceptance [17]. Therefore, there is a clear need for an integrated framework that combines accurate anomaly detection with efficient, real-time interpretability in medical settings [18].

II. AIM AND OBJECTIVE

A. Aim

The aim of this research is to design, implement, and evaluate an explainable anomaly detection framework for mIoT telemetry. It seeks to address the “interpretability gap” by combining multiple detection models with XAI techniques, providing clinicians with clear, feature-level explanations to improve trust and reduce alarm fatigue.

B. Objectives

- **Literature Analysis:** Critically review anomaly detection methods and XAI techniques, focusing on their suitability for high-frequency physiological data.
- **Data Preparation:** Preprocess mIoT data using cleaning, imputation, and feature engineering (e.g., Mean Arterial Pressure) to handle noise and variability.
- **Model Development:** Implement and benchmark Isolation Forest, KNN, LOF, and Autoencoder models using metrics suited for imbalanced data, including F1-score and MCC.
- **XAI Integration:** Apply SHAP for global explanations and LIME for local interpretability to explain anomaly predictions.
- **Performance Trade-off Analysis:** Evaluate the balance between model complexity, computational efficiency, and interpretability for real-time feasibility.

C. Research Contribution

This research contributes to digital healthcare by developing a hybrid anomaly detection framework that separates detection

performance from explanation generation. The key contributions are:

- **Multi-Model Benchmarking:** Evaluation of Isolation Forest, KNN, LOF, and Autoencoder models on mIoT data, showing that tree-based methods can outperform deep learning in noisy, small-scale settings.
- **Dual-Layer Explainability:** Integration of SHAP for global feature importance and LIME for local explanations, enabling both overall and instance-level interpretability.
- **Domain-Specific Feature Engineering:** Creation of clinically relevant features (e.g., Mean Arterial Pressure and device stability metrics) to improve detection of both physiological anomalies and sensor-related faults.

III. LITERATURE REVIEW

A. Evolution of Anomaly Detection in Medical IoT (mIoT)

Traditional rule-based systems struggle with the non-stationary nature of physiological signals [1]. Consequently, research has shifted toward unsupervised machine learning. Isolation Forest (IF) is a common baseline due to its efficient recursive partitioning [13], while LSTM Autoencoders are increasingly used to model temporal dependencies [10]. However, the complexity of deep learning introduces a "transparency deficit," hindering clinical adoption [11].

B. The "Interpretability Gap" and the Emergence of XAI

The "black box" nature of mIoT models reduces clinical trust [17]. Explainable AI (XAI) addresses this through model transparency. SHAP, rooted in game theory, provides consistent global feature attribution to identify key variables like SpO_2 and Heart Rate [23]. Conversely, LIME offers local explanations for individual predictions, justifying patient-level clinical decisions [25].

C. The Computational Trilemma: Edge-Based Explainability

Deploying XAI in mIoT face a trade-off between accuracy, interpretability, and computational cost. While SHAP offers detailed insights, its overhead often exceeds the capacity of resource-constrained edge devices [4]. This research evaluates the feasibility of real-time explainability by comparing lightweight models like Isolation Forest against complex neural networks.

IV. RESEARCH QUESTIONS

- 1) To what extent can unsupervised anomaly detection paradigms specifically tree-based isolation vs. deep reconstruction effectively identify physiological abnormalities in high-frequency mIoT sensor data?
- 2) How do post-hoc interpretability frameworks (SHAP and LIME) compare in providing clinically coherent justifications for anomaly alerts across diverse machine learning architectures?
- 3) What are the computational and performance trade-offs when integrating complex XAI methods into an mIoT monitoring pipeline, and how do these impact the feasibility of real-time edge deployment?

V. DATA SET SELECTION AND JUSTIFICATION

Dataset Description: This study utilises the Secure Healthcare IoT Monitoring Dataset [9], which provides multi-dimensional telemetry representative of real-world mIoT environments. Unlike traditional datasets focused solely on biometrics, it combines physiological signals such as Heart Rate (bpm), Body Temperature ($^{\circ}C$), and Blood Pressure (mmHg) with device-level metadata including Battery Level and Signal Strength.

Dataset link: <https://iotdataset.com/data/secure-healthcare-iot-monitoring-anomaly-detection-patient-safety>

Justification:

- **Dual-Perspective Analysis:** The integration of physiological and device data enables differentiation between clinical anomalies and technical faults (e.g., sensor noise due to low battery) [11], [12].
- **XAI Compatibility:** The structured tabular format supports SHAP and LIME, allowing clear feature-level explanations (e.g., " SpO_2 drop") that improve clinical interpretability [23], [13].
- **Feature Complexity:** Inclusion of network-level attributes (e.g., access source) creates a richer feature space, supporting detection of multi-dimensional anomalies [25].

Limitations:

- **Synthetic Characteristics:** The dataset may not fully reflect real-world clinical variability, potentially inflating performance results, [16].
- **Class Imbalance:** The skewed distribution of anomalies necessitates the use of F1-score and MCC over accuracy for reliable evaluation [22], [14].
- **Temporal Constraints:** Irregular sampling limits the use of sequence-based models (e.g., RNNs), justifying the selection of static models such as Isolation Forest and Autoencoders [13], [21].

VI. RESEARCH METHODOLOGY

A. Research Design

This study adopts a modular, reproducible pipeline to ensure analytical robustness and clinical interpretability, aligned with real-world mIoT data processing.

The workflow consists of six stages:

- **Data Ingestion:** Collection of mIoT telemetry data ($N = 2000$) including physiological, device, and network attributes.
- **Preprocessing:** Cleaning data using bidirectional imputation and removal of physiologically implausible outliers.
- **Feature Engineering:** Transforming raw inputs into clinically meaningful features, including derived metrics such as Mean Arterial Pressure (MAP).
- **Normalization:** Applying Z-score standardisation to ensure consistent feature scaling and improve model performance.

- **Model Implementation:** Training multiple anomaly detection models (Isolation Forest, KNN, LOF, Autoencoder) to capture diverse anomaly patterns.
- **Explainability Layer:** Integrating SHAP and LIME to provide global and local interpretations of model predictions.

B. Data Preprocessing

Beyond standard preprocessing techniques, this study incorporates domain-specific transformations to enhance the sensitivity and clinical relevance of anomaly detection.

Physiological Feature Synthesis: The “Blood Pressure” attribute, originally represented as a string, was decomposed into systolic and diastolic components. To derive a more clinically meaningful representation of cardiovascular stability, Mean Arterial Pressure (MAP) was computed using:

$$MAP \approx \frac{1}{3}BP_{systolic} + \frac{2}{3}BP_{diastolic}$$

This transformation enables the model to capture overall blood flow dynamics rather than relying on isolated readings.

BP_Systolic	BP_Diastolic	BP_MAP
116	84	94.666667
114	75	88.000000
141	99	113.000000
114	67	82.666667
114	77	89.333333

TABLE I

DERIVED BLOOD PRESSURE FEATURES INCLUDING MEAN ARTERIAL PRESSURE (MAP)

Scaling Justification: Given the heterogeneous nature of physiological features, normalization was essential. Standard-Scaler was applied to transform all features to a common scale ($\mu = 0, \sigma = 1$). This step is particularly critical for distance-based models such as KNN and LOF, where feature magnitude directly influences anomaly detection.

C. Feature Engineering

The feature engineering phase was bifurcated into two domains to help the model distinguish between medical emergencies and technical malfunctions:

Clinical Domain: Decomposing complex vitals and synthesizing MAP to improve the detection of physiological deterioration.

Operational Domain (Metadata): Integrating variables such as Battery Level and Signal Strength. By including these, the XAI layer (SHAP/LIME) can identify if an anomaly was triggered by a clinical crisis or a hardware failure (e.g., sensor noise induced by voltage drops) [10].

$$MAP = \frac{Systolic + 2(Diastolic)}{3}$$

This transformation provides a smoothed representation of cardiovascular stability and improves the model’s sensitivity to circulatory anomalies.

1) **Device and Operational Metadata:** To address the interpretability gap specifically distinguishing physiological anomalies from technical artifacts device-level features were incorporated [28].

- **Battery and Signal Dynamics:** Features such as Battery Level and Signal Strength enable SHAP/LIME to associate degraded device conditions with sensor noise or unstable readings [30].
- **Access Pattern Encoding:** Access sources (e.g., App vs. Web) were label encoded to detect abnormal usage patterns and potential cyber-physical anomalies [7].

2) **Data Transformation and Scaling:** Distance-based models such as KNN and LOF are sensitive to feature magnitude [18]. To prevent dominant influence from high-range variables (e.g., Heart Rate), Z-score normalization was applied, standardising features to mean ($\mu = 0$) and standard deviation ($\sigma = 1$) [14]:

$$z = \frac{x - \mu}{\sigma}$$

D. Algorithmic Functionality

Four anomaly detection models were implemented to capture diverse patterns:

- **Isolation Forest (IF):** A tree-based model that isolates anomalies via random partitioning. Anomalies have shorter path lengths due to their rarity [9].
- **K-Nearest Neighbors (KNN):** A distance-based method where anomalies are identified by large distances from neighbouring points. Performance depends on proper feature scaling [18].
- **Local Outlier Factor (LOF):** A density-based approach that detects anomalies by comparing local density with neighbouring points, effective for contextual anomalies [18].
- **Autoencoder:** A neural network that learns normal patterns and detects anomalies via reconstruction error, measured using Mean Squared Error (MSE) [10]:

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

Samples exceeding a threshold ($\mu + 2\sigma$) are classified as anomalies.

E. Explainable AI (XAI) Implementation

To address the “black-box” nature of the models, a dual-layer XAI framework was implemented:

- **SHAP (Global Interpretability):** Based on cooperative game theory, SHAP assigns each feature a Shapley value representing its contribution to model predictions across all feature combinations [23], [12]. This provides a consistent global ranking of feature importance.
- **LIME (Local Interpretability):** LIME explains individual predictions by approximating the model locally with a simple surrogate model, enabling identification of whether anomalies are driven by physiological changes

(e.g., SpO_2 drop) or technical issues (e.g., battery failure) [25], [30].

- **Clinical Relevance:** Combining SHAP and LIME enables both global insight and instance-level explanation, supporting clinical decision-making and reducing alarm fatigue caused by non-interpretable alerts [11],[8], [9], [24].

F. Evaluation Criteria

Anomaly detection is inherently affected by severe class imbalance, where normal events dominate anomalous cases [15]. Consequently, accuracy is not a reliable performance indicator [18],[22]. This study therefore adopts a multi-metric evaluation framework to ensure robust and clinically meaningful assessment.

The following metrics are used:

- **Precision**
- **Recall**
- **F1-Score**
- **AUC-ROC**

In addition, the Matthews Correlation Coefficient (MCC) is employed due to its robustness under imbalanced conditions [22]. It is defined as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Justification: Unlike the F1-score, MCC incorporates all confusion matrix elements to provide a balanced correlation [22], ensuring high-stakes medical anomalies are only validated when all classification components including True Negatives are optimized.

VII. FINDINGS AND DISCUSSION

A. Descriptive Analysis

The mIoT dataset ($N = 2000$) exhibits high variance: heart rate averages $\mu = 85.2$ bpm ($\sigma = 15.4$), and blood pressure shows a long-tail distribution indicative of rare clinical events. With anomalies at only 13.8%, evaluation prioritizes the Matthews Correlation Coefficient (MCC) and F1-score [22] over accuracy, which is misleading in sparse medical contexts.

B. Data Visualization

1) *Feature Correlation Heatmap:* A strong correlation ($r > 0.92$) between Systolic Blood Pressure and Mean Arterial Pressure (MAP) validates MAP as a stable cardiovascular indicator [20]. In contrast, weak correlations between physiological vitals and device metadata (Battery/Signal) necessitate a hybrid analytical approach to distinguish biological crises from hardware artifacts [29], [30].

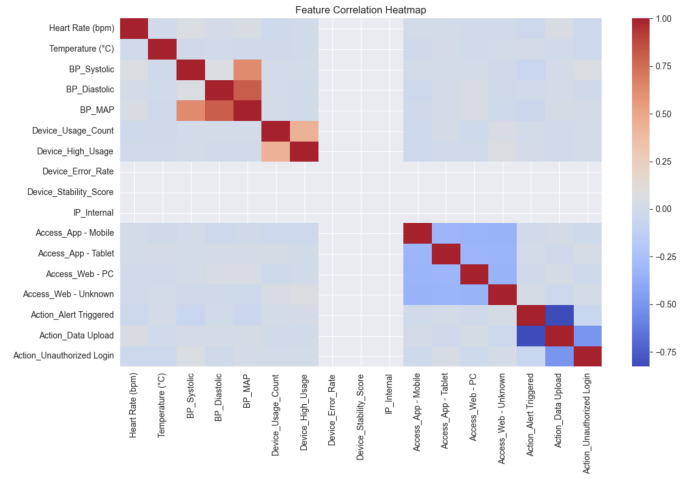


Fig. 1. Feature correlation heatmap illustrating relationships between physiological, device, and behavioural attributes in the mIoT dataset.

2) **Model 1 Isolation Forest:** IF outperformed other models (AUC-ROC: 0.9788). By isolating anomalies via recursive partitioning, it effectively identified physiological clusters like concurrent hypertension and bradycardia, which signify circulatory instability [9],[13]. Its computational efficiency supports real-time deployment in mIoT gateways [4].

3) **Model 2 KNN:** KNN identifies outliers using the Euclidean metric:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

While effective for extreme outliers, KNN struggles with overlapping "normal" and "abnormal" regions. High computational costs for distance calculations also limit its real-time utility.

4) **Model 3 LOF:** LOF measures local reachability density to find contextual anomalies [18]. In this study, it performed poorly (AUC-ROC: 0.5037), as mIoT stochastic noise likely obscured density gradients. While useful for patient-specific variations, it proved ineffective against global noise.

5) **Model 4 Auto Encoder:** The Autoencoder identifies anomalies through reconstruction error ($RE = |x - \hat{x}|^2$) [10]. However, a low recall (0.0545) indicates it overfitted the "normal" manifold, absorbing physiological noise as expected signal rather than flagging it. This conservative performance makes it unreliable for capturing true positives in noisy datasets.

C. Comparing models

Table 3 reveals a distinct performance divergence across models under 13.8 percent class imbalance [15]. Given this sparsity, Recall, AUC-ROC, and F1-score provide more reliable interpretations than accuracy [22].

TABLE II
EVALUATION METRICS ON TEST SET

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Isolation Forest	0.9075	0.5978	1.0000	0.7483	0.9788
KNN	0.9125	0.6190	0.9455	0.7482	0.9612
LOF	0.8575	0.0000	0.0000	0.0000	0.5037
Autoencoder	0.8700	1.0000	0.0545	0.1034	0.9306

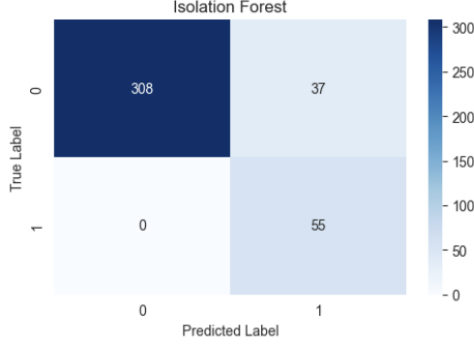


Fig. 2. Confusion matrix for Isolation Forest model showing classification performance on anomaly detection task.

Isolation Forest (IF) demonstrates superior robustness, as evidenced by an AUC-ROC of 0.9788 and perfect recall (1.0000) in Fig. 2. This sensitivity is vital in clinical settings to prevent missed deterioration signals [3], confirming IF’s suitability for high-variance mIoT data [13].

KNN (see Fig. 3) shows competitive accuracy (0.9125) and recall (0.9455). However, its reliance on Euclidean distance limits interpretability in heterogeneous feature spaces where scaling often distorts results [18].

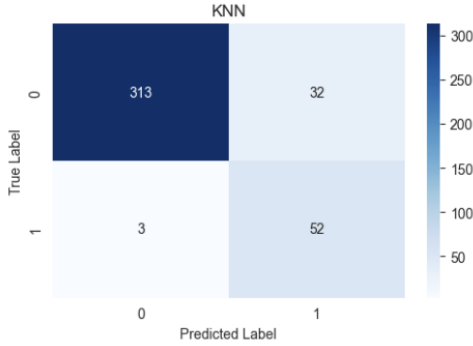


Fig. 3. Confusion matrix for K-Nearest Neighbors (KNN) model showing classification performance on anomaly detection task.

Autoencoder performance, illustrated in Fig. 5, reveals a significant failure mode: while achieving perfect precision, its critical recall (0.0545) reflects extreme conservatism. This suggests the model overfits ”normal” patterns, failing to detect subtle physiological deviations essential for early-warning systems [10].

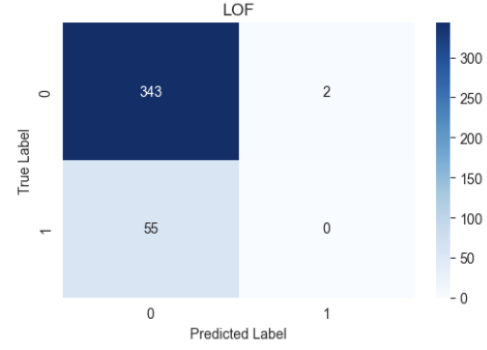


Fig. 4. Confusion matrix for Local Outlier Factor (LOF) model showing poor classification performance under high-dimensional noisy data.

LOF (see Fig. 4) yielded the weakest results (AUC-ROC: 0.5037), indicating instability in local density estimation due to overlapping distributions and noise.

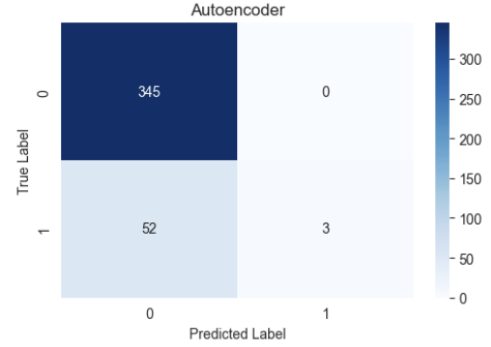


Fig. 5. Confusion matrix for Autoencoder model showing high precision but extremely low recall due to conservative anomaly detection behaviour.

D. SHAP

SHAP utilizes Cooperative Game Theory to quantify feature contributions via the Shapley value (ϕ_i), representing the average marginal impact across all possible coalitions [23]:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{i\}) - f_x(S)]$$

In mIoT, this prevents ”feature hijacking” by dominant variables [11].

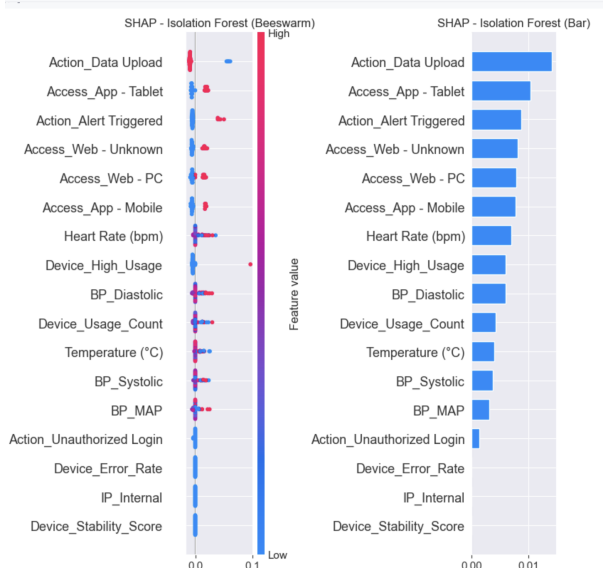


Fig. 6. SHAP summary plot (beeswarm) for Isolation Forest showing global feature importance for anomaly detection.

The beeswarm plot in Fig. 6 identifies Heart Rate as the primary global driver, while MAP acts as a contextual modifier, triggering high-confidence alerts only when coupled with other stressors.

E. LIME

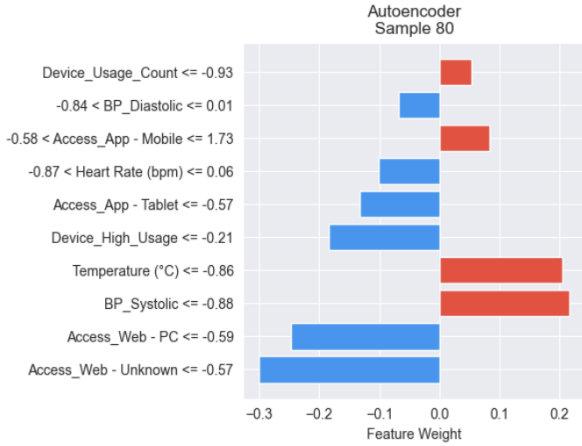


Fig. 7. LIME local explanation for Autoencoder output (Sample 80), highlighting feature contributions responsible for anomaly classification at instance level.

LIME provides local interpretability by fitting a surrogate linear model g to perturbations around a specific instance x [13]:

$$\xi(x) = \arg \min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

where L measures local fidelity and $\Omega(g)$ penalizes complexity [19]. In the instance-level analysis shown in Fig. 7, LIME identifies Battery Level as the primary driver for an alert. This

distinguishes device-layer artifacts from biological deterioration [28], [30], a critical capability for reducing clinician alarm fatigue and preventing unnecessary interventions [8], [9], [17].

F. Metrics

Given the class imbalance in mIoT data, accuracy is an unreliable indicator; therefore, evaluation relies on Precision, Recall, and the F1-score [15], [22].

Precision quantifies the proportion of correctly identified anomalies, crucial for minimizing unnecessary clinical interventions. Recall measures the proportion of actual anomalies detected and is the most critical clinical metric, as false negatives directly result in missed patient deterioration [3], [8].

The F1-score provides a harmonic balance between the two, offering a consolidated measure of effectiveness under imbalanced conditions [22]. Ultimately, recall is prioritized due to the high clinical cost of missed anomalies compared to false alerts [9], [17].

Overall, recall is prioritized in this study due to the higher cost of missed anomalies compared to false alerts in medical monitoring systems [9], [17].

VIII. RESULTS AND DISCUSSION

Isolation Forest (IF) emerged as the superior model, achieving a peak AUC-ROC (0.9788) and perfect recall (1.0000). Its recursive partitioning structure effectively isolates anomalous events [13], meeting the critical clinical requirement for high sensitivity to prevent missed deterioration [3].

Conversely, LOF performed poorly (AUC-ROC: 0.5037), signaling instability in local density estimation due to the stochastic noise inherent in mIoT streams. Despite its theoretical capacity, the Autoencoder underperformed due to noise sensitivity and data scarcity [10]. While training reached a final loss of 0.2095, the model overfitted the "normal" physiological manifold, creating an overly narrow reconstruction boundary ($threshold = 0.6829$).

Consequently, it only flagged extreme outliers while missing subtle early-stage anomalies, resulting in high precision but critically low recall a profile unsuitable for early-warning clinical monitoring.

Epoch [10/50] Loss: 0.442269
 Epoch [20/50] Loss: 0.307068
 Epoch [30/50] Loss: 0.256361
 Epoch [40/50] Loss: 0.232090
 Epoch [50/50] Loss: 0.209541

Training anomalies detected: 18/1600
 Autoencoder threshold: 0.682924

A. Accuracy vs Interpretability

A critical finding of this study is the trade-off between model complexity and interpretability:

- Autoencoder (Deep Learning): Offers high representational capacity but lacks transparency. Its opaque decision

process and sensitivity to noisy telemetry limit clinical trust.

- Isolation Forest (Tree-based): Balances robust performance with inherent interpretability via feature-based reasoning and XAI compatibility.
- KNN LOF (Distance/Density-based): While theoretically interpretable, these models suffer from instability in high-dimensional, noisy medical datasets.

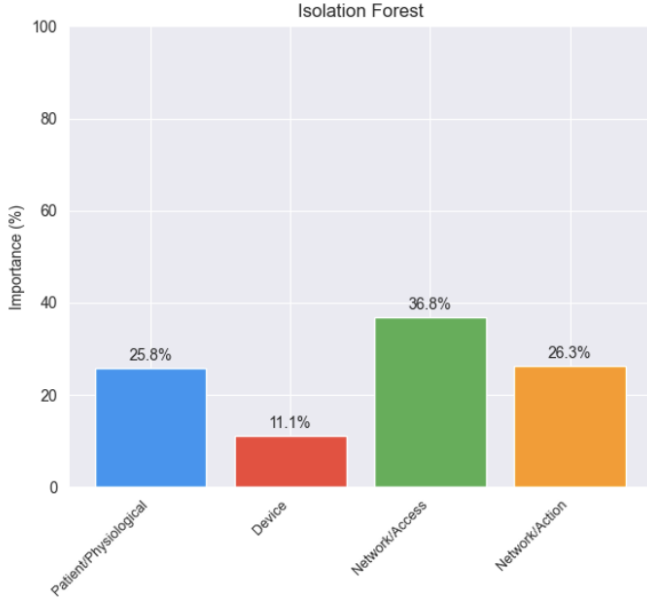


Fig. 8. Feature Importance Device vs Patient Isolation Forest Model

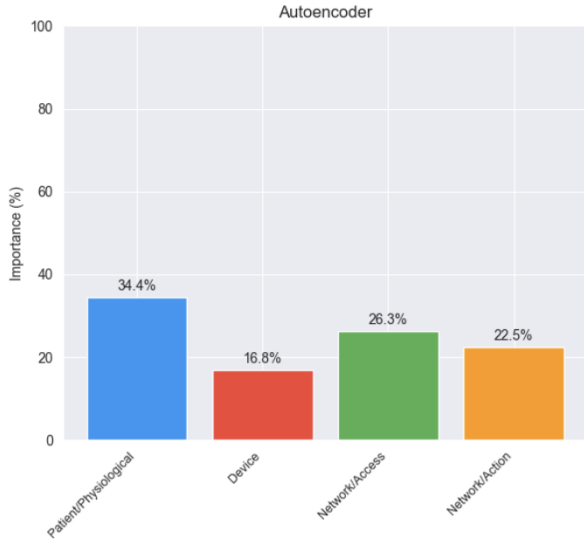


Fig. 9. Feature Importance Device vs Patient AutoEncoder Model performance

Global feature importance analysis using SHAP reveals that while physiological vitals (Heart Rate and MAP) are the primary drivers for high-confidence alerts, device-layer metadata

(Battery/Signal) frequently contributes to noise-driven anomalies. This confirms that model complexity does not equate to clinical utility; interpretability is a functional requirement for medical trust and adoption [4], [17].

B. Importance of XAI in This Study:

Explainable AI bridges the gap between predictive performance and clinical usability [15], [19]. In this study, SHAP and LIME transform model outputs into meaningful explanations [23], [25]. For example, in Case 142, LIME identifies battery degradation rather than physiological abnormality as the primary cause of the anomaly [28], [30]. Without this layer, such an alert could exacerbate "alarm fatigue" [9]. XAI does not improve model accuracy, but it significantly improves clinical decision quality.

- SHAP (Global level) identifies consistent physiological drivers such as Heart Rate and MAP, confirming that the model aligns with known medical indicators of cardiovascular instability.

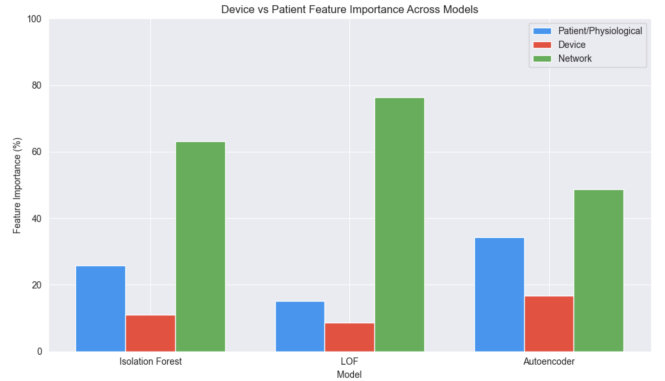


Fig. 10. Device vs Patient Feature Importance across models SHAP

- LIME (Local level) provides instance-specific explanations, allowing clinicians to understand why a particular alert was triggered.

1) *Key Insight:* XAI does not improve model accuracy, but it significantly improves clinical decision quality, which is equally important in healthcare systems.

Therefore, SHAP and LIME are not auxiliary tools but essential components of safe mIoT deployment, particularly in reducing alarm fatigue and improving clinician trust.

IX. CONCLUSION

This research successfully developed an explainable anomaly detection framework, directly answering the study's core objectives:

- Resolution of the Interpretability Gap: By integrating SHAP and LIME, the study proved that black-box outputs can be translated into clinically actionable justifications.
- Optimization of Clinical Detection: Findings identify Isolation Forest as the superior model for noisy mIoT datasets, offering the best balance of high recall and computational efficiency.

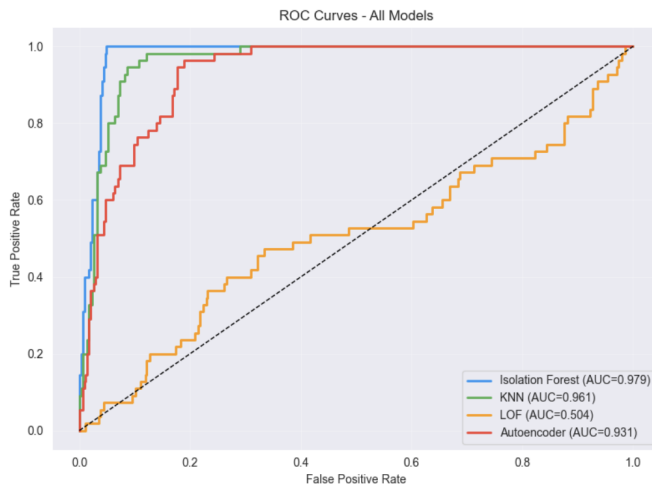


Fig. 11. ROC

- Validation of Domain-Specific Engineering: The synthesis of MAP was confirmed as a critical factor in improving physiological interpretability.
- Mitigation of Alarm Fatigue: Local explanations (LIME) demonstrated the ability to differentiate sensor artifacts from genuine medical emergencies.

Ultimately, this study confirms that explainability is not merely an auxiliary feature but a fundamental requirement for the safe deployment of AI in healthcare. The proposed framework provides a scalable foundation for future patient monitoring systems where clinician trust and patient safety are paramount. The findings confirm that explainability is not optional in healthcare AI systems but a fundamental requirement for safe deployment.

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